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Single-cell multiomics analysis reveals CTCF as a key regulator of lung morphogenesis and progenitor maintenance

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Lung development generates a complex tree-like architecture through proximal-distal patterning and branching morphogenesis. However, the gene regulatory programs governing embryonic lung development remain poorly understood. Here, we present a comprehensive single-cell multi-omics atlas of mouse embryonic lungs, integrating gene expression and chromatin accessibility profiles. Through systematic analysis, we identify 13 distinct cell types and map cis-regulatory elements, peak-to-gene linkages, and transcription factors underlying lung development. Leveraging this multi-modal dataset, we uncover lineage-determining transcription factors driving cell differentiation, including the Activated Protein-1 complex. We further delineate gene regulatory networks involving diverse transcription regulators, including CCCTC-binding factor (CTCF). Using the *Ctcf* conditional knockout mouse, coupled with histological and multi-omics analyses, we demonstrate that CTCF orchestrates lung progenitor maintenance and branching morphogenesis by modulating both gene expression and chromatin accessibility. Thus, our study provides a multi-omics resource and mechanistic insights for transcriptional regulation of lung morphogenesis.

During its development, the lung progresses through five morphologically distinct yet partially overlapping stages: embryonic, pseudoglandular, canalicular, saccular, and alveolar[1–3]. Disruptions in prenatal lung development, caused by genetic abnormalities, preterm birth, amniotic fluid leakage, or intrauterine infections can result in congenital lung defects such as congenital pulmonary airway malformation (CPAM) and bronchopulmonary dysplasia (BPD)[4,5]. Although the lung possesses regenerative capacity, most prenatal defects are irreparable postnatally, potentially causing severe respiratory disorders or even death. Consequently, elucidating the regulatory programs governing lung morphogenesis is critical for advancing our understanding of lung disease pathogenesis and developing novel therapeutic interventions.

Deciphering the mechanisms of cell lineage determination during lung morphogenesis has been a major research focus. It has been known that lung endoderm arises from Nkx2-1-expressing cells in the

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ventral anterior foregut[6]. These Nkx2-1+ early endoderm progenitors give rise to the entire lung epithelium, organizing into a proximal-distal pattern via branching morphogenesis[7]. During this process, Sox9+ epithelial progenitors at the branching tips function as lung endoderm stem cells, exhibiting bidirectional differentiation potential. They generate proximal Sox2+ progenitors for airway formation and distal Sox9+ progenitors for alveolar development[8–10]. Proximal Sox2+ progenitors further differentiate into ciliated and secretory cells, while distal Sox9+ progenitors produce alveolar type 1 (AT1) and type 2 (AT2) cells, crucial for gas exchange and surfactant secretion[11]. In addition to epithelial cells, mesenchymal cells represent a major cell population in embryonic lungs, undergoing complex lineage specification to generate diverse pulmonary mesenchymal cell types[12].

Lung branching and early cell fate specification are tightly controlled by extrinsic and intrinsic factors[13,14]. Mesenchymal cells secret morphogens such as FGF10, WNT2/2b, and BMP4, which orchestrate epithelial proliferation, differentiation, and migration[15–18]. Epithelial cells produce WNT7b and SHH, which modulate mesenchymal development and function, establishing a reciprocal loop[19–21]. Key transcription factors (TFs), including SOX9, ID2, SOX2, ETV5, and FOXA1/2, have been implicated in regulating lung branching morphogenesis[22–26]. Disruption of epigenetic regulators, such as miRNA regulator DICER, histone deacetylases (HDAC1/2/3), DNA methyltransferase 1 (DNMT1), and chromatin remodeler ZNHIT1 and ARID1A, leads to diverse lung phenotypes, ranging from branching defects to impaired cell specification or maintenance, highlighting the critical role of epigenetic mechanisms in lung development[27–32]. Additionally, organelles like mitochondria and centrioles play essential roles in proximal-distal patterning and lung progenitor maintenance[33,34]. To fully unravel the complexities of lung morphogenesis, it is imperative to systematically decode how these extrinsic and intrinsic regulatory events integrate with chromatin dynamics[35].

Lineage commitment is governed by cell type-specific gene expression, precisely regulated by cis-regulatory elements (CREs) and trans-acting factors (TAFs)[36,37]. Recent advances in single-cell technologies have revolutionized our ability to dissect this regulatory mechanism[38]. Tools such as single-cell RNA sequencing (scRNA-seq) and single-cell assay for transposase-accessible chromatin using sequencing (scATAC-seq) now allow for investigating transcriptional and epigenetic dynamics at unprecedented single-cell resolution. These approaches have been instrumental in uncovering cellular heterogeneity and gene expression patterns during lung morphogenesis[39–43]. Nevertheless, the epigenomic states and gene regulatory networks (GRNs) that orchestrate lung development remain largely unclear.

In this study, we employed paired single-nuclear multi-omics technologies (snRNA/ATAC-seq) and cleavage under targets and tagmentation (CUT&Tag) to dissect the gene regulatory programs underlying embryonic lung development. By constructing a single-cell atlas of gene expression and chromatin accessibility in mouse fetal lungs, we systematically delineated cell type-specific transcriptional and epigenetic dynamics during lung morphogenesis. We identified key regulators, such as the Activated Protein-1 (AP-1) complex, driving epithelial and mesenchymal differentiation trajectories. Leveraging enhancer-driven GRNs, we further pinpointed CCCTC-binding factor (CTCF) as a critical regulator of lung progenitor maintenance and branching morphogenesis. These findings provide mechanistic insights into how chromatin regulation orchestrates lineage-specific gene expression during lung morphogenesis.

Results

scRNA-seq analysis of early lung development

To study early lung branching morphogenesis, we performed scRNA-seq on embryonic mouse lungs, analyzing two biological replicates at each developmental time point [embryonic day 11.5 (E11.5) and E12.5] (Fig. 1a). This approach yielded 30,250 high-quality single-cell transcriptomes (Fig. 1b and Supplementary Fig. 1a). According to the expression of known marker genes[44], visualization using uniform manifold approximation and projection (UMAP) revealed that the single-cell atlas comprised multiple distinct cell types: epithelial cells (*Epcam*, *Nkx2-1*), mesenchymal cells (*Wnt2*, *Pdgfrb*), mesothelium (*Upk3b*, *Wt1*), endothelial cells (*Ramp2*, *Cd34*), neuroendocrine cells (*Fabp7*, *Ascl1*), and immune cells (*Apoe*, *Ptprc*) (Supplementary Fig. 1b, c).

Within the epithelial compartment, we identified two major progenitor populations: Sox9+ distal progenitors and Sox2+ proximal progenitors (Fig. 1c–h and Supplementary Data 1, 2). The mesenchymal compartment was further subdivided into specialized subsets, including proliferating Wnt2+ mesenchymal progenitors, Wnt2+ mesenchymal progenitors, Acta2+ airway smooth muscle (ASM) progenitors, intermediate mesenchymal cells, Sfrp2+ adventitial fibroblasts, Ebf1+ mesenchymal cells, Sox9+ mesenchymal cells (tracheal mesenchyme)[45], and Acta2+Pdgfrb+ vascular smooth muscle (VSM) cells (Fig. 1i–n and Supplementary Data 3, 4). We also noted that the composition of mesenchymal cells was dynamic: a subset of Ebf1+ mesenchymal cells was detectable at E11.5, whereas these cells became nearly undetectable by E12.5. Conversely, Acta2+Pdgfrb+ VSM cells emerged as detectable at E12.5, albeit at a relatively low proportion. Pseudotime trajectory analysis of E12.5 mesenchymal cells revealed that Wnt2+ mesenchymal cells functioned as progenitor cells, giving rise to Acta2+ ASM, Acta2+Pdgfrb+ VSM, and Sox9+ mesenchymal cells through three distinct developmental trajectories (Supplementary Fig. 2).

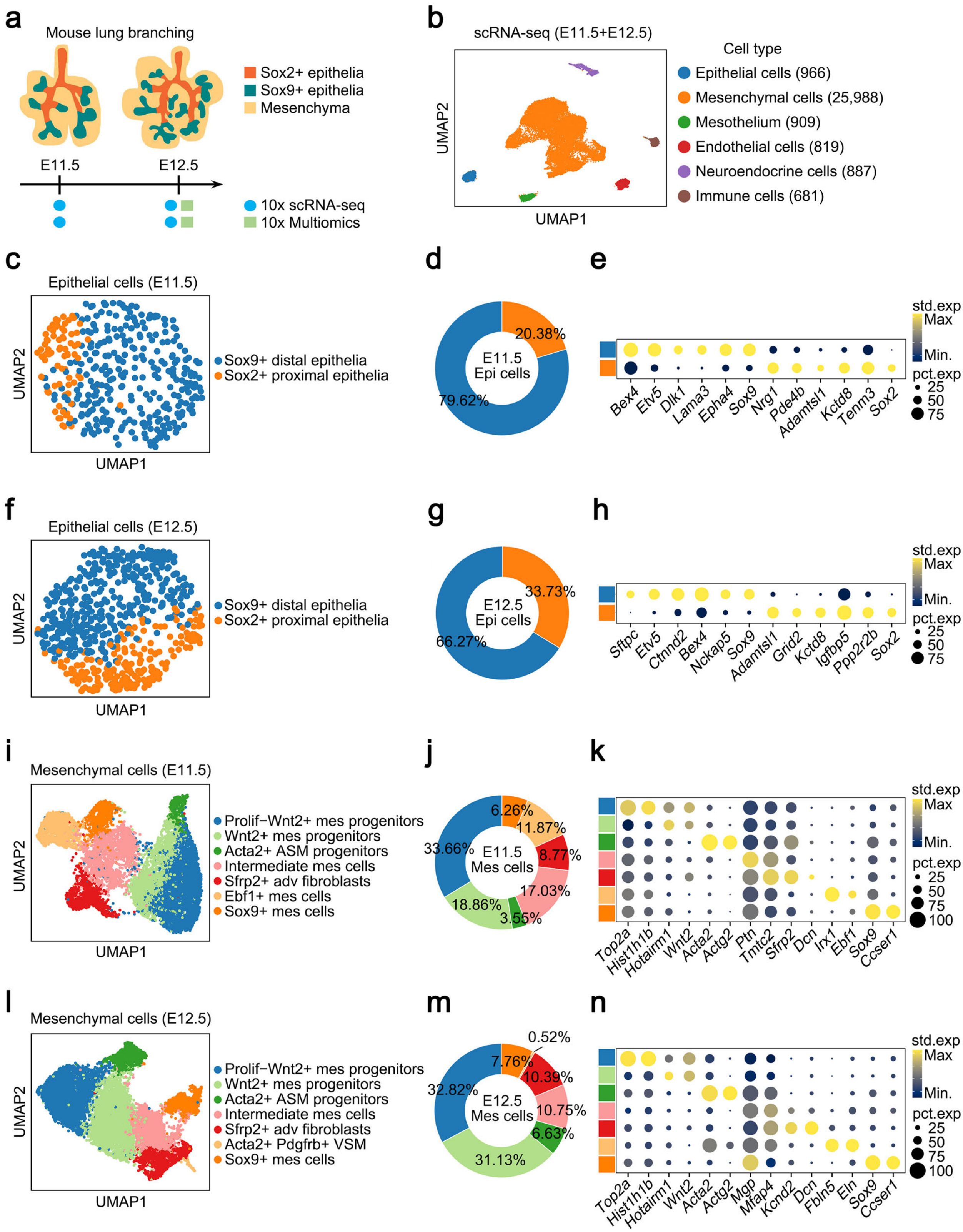
Together, we constructed a comprehensive single-cell transcriptome atlas of early lung development, allowing further exploration of cellular composition and gene expression profiles during lung branching morphogenesis.

Single-cell multiomics profiling of mouse lung branching

To simultaneously investigate cellular heterogeneity, gene expression, and chromatin dynamics during lung branching, we performed multiomics profiling of nuclei isolated from E12.5 lungs, a critical stage for branching morphogenesis[2]. To enrich epithelial cells in the dataset, we sorted and supplemented epithelial cells into the whole lung cell population (Supplementary Fig. 3a and methods). Using 10x Genomics paired snRNA-seq and snATAC-seq, we simultaneously captured gene expression and chromatin accessibility profiles (Fig. 2a). To ensure consistent data quality across all replicates, we applied a set of key quality control criteria to each replicate (Supplementary Fig. 3b–d). Totally, we retained 15,890 fetal mouse lung cell nuclei for downstream analysis.

Next, we integrated reduced-dimensionality information from snRNA-seq and snATAC-seq using the weighted nearest-neighbor (WNN) method, identifying 13 distinct cell types. These included epithelial cells (Sox9+ distal epithelia and Sox2+ proximal epithelia), mesenchymal cells (proliferating Wnt2+ mesenchymal progenitors, Wnt2+ mesenchymal progenitors, Acta2+ ASM progenitors, intermediate mesenchymal cells, Sfrp2+ adventitial fibroblasts, Acta2+Pdgfrb+ VSMs, Sox9+ mesenchymal cells, and mesothelial cells), endothelial cells, neuroendocrine cells, and immune cells (Fig. 2b and Supplementary Fig. 4a). A total of 4295 cell-type-specific genes (marker genes) were identified across distinct cell clusters (Supplementary Fig. 4b, c and Supplementary Data 5). Gene ontology (GO) term analysis revealed that these marker genes were enriched in known cellular functions, such as ‘epithelial tube morphogenesis’ (Sox9+ distal epithelia), ‘muscle cell differentiation’ (Acta2+ ASM progenitors), ‘muscle system process’ (Acta2+Pdgfrb+ VSMs), and ‘regulation of vasculature development’ (endothelial cells) (Supplementary Fig. 4b).

To explore the relationships between gene expression and chromatin accessibility, we calculated gene activity scores for marker genes using snATAC-seq data within the same cells. Our analysis revealed a



strong positive correlation between marker gene expression and chromatin accessibility (Fig. 2c, d). We identified a total of 251,078 open chromatin sites, referred to as CREs (Supplementary Data 6). TF motif analysis of CREs revealed the most enrichment for chromatin regulators, such as CTCF (Supplementary Fig. 5a, b).

We further identified 42,747 differentially accessible chromatin regions (DARs) across distinct cell clusters (Supplementary Data 7). Genomic annotation of these DARs indicated that the majority localized in promoters, introns, and intergenic regions (Fig. 2e). TF motif analysis of cell-type enriched DARs highlighted enrichment for lineage-determining TFs, including NKX2-8 and FOXA2 in embryonic lung epithelia[46], and ETV1 and ETV4 in endothelial cells[47] (Fig. 2f). These results were further validated by TF motif scores (calculated using chromVAR) and TF footprint analyses (Fig. 2g and Supplementary Fig. 5c).

Fig. 1 | Single-cell RNA-seq analysis of mouse early lung development. **a** Schematic representation of the experimental design. **b** UMAP visualization of scRNA-seq, colored by 6 annotated cell types in mouse embryonic lung at E11.5 and E12.5. **c** UMAP visualization of E11.5 epithelial cells in scRNA-seq datasets. **d** Pie plot showing the proportion of different E11.5 epithelial cell types. **e** Dot plot showing the expression of marker genes across various E11.5 epithelial cell clusters. **f** UMAP visualization of E12.5 epithelial cells in scRNA-seq datasets. **g** Pie plot showing the proportion of different E12.5 epithelial cell types. **h** Dot plot showing the expression of marker genes across various E12.5 epithelial cell clusters. **i** UMAP visualization of E11.5 mesenchymal cells in scRNA-seq datasets. **j** Pie plot showing the proportion of different E11.5 mesenchymal cell types. **k** Dot plot showing the expression of marker genes across various E11.5 mesenchymal cell clusters. **l** UMAP visualization of E12.5 mesenchymal cells in scRNA-seq datasets. **m** Pie plot showing the proportion of different E12.5 mesenchymal cell types. **n** Dot plot showing the expression of marker genes across various E12.5 mesenchymal cell clusters.

Collectively, we construct a single-cell atlas of gene expression and chromatin accessibility during mouse lung branching.

Highly regulated genes define cellular identity

To study the chromatin regulation underlying cell-type-specific gene expression, we computed direct connections between gene expression and regulatory elements, termed peak-to-gene linkages (Fig. 3a). This analysis identified 29,848 positively correlated linkages and 767 negatively correlated ones (Supplementary Data 8). Among these, positive peak-gene pairs connected 21,346 ATAC peaks to 5290 genes, most of which were marker genes. To characterize these linkages, we performed k-means clustering based on the expression levels of linked genes, organizing them into 11 distinct modules (Fig. 3b). GO term enrichment analysis revealed that each module of genes was associated with cell-type-specific functions (Fig. 3b, right). For instance, functions of genes within C1 and C2 modules were enriched in epithelial morphogenesis and branching development, aligning the roles of lung epithelial cells. These results suggest that co-variation of positively correlated linkages reliably distinguishes each cell type.

Next, we classified the above genes based on the number of ATAC peaks linked to each gene. Genes with more than 10 peaks were designated as highly regulated genes (HRGs; 895 genes), while those with 10 or fewer peaks were classified as lowly regulated genes (LRGs; 4395 genes) (Fig. 3c). We next categorized the marker genes of different cell types based on whether they are classified as HRGs or LRGs. Comparative analysis revealed two key observations: the overall expression levels of HRGs were significantly higher than those of LRGs, and a greater proportion of cell-type marker genes fell into the highly regulated category rather than the lowly regulated one (Fig. 3d). Specifically, cell types in which highly regulated marker genes were enriched included endothelial cells (35.2%), Sox9$^+$ distal epithelia (25.8%), and immune cells (24.0%). Key HRGs included *Nkx2-1*, *Sox9*, *Sox2*, *Cdh1*, and *Tbx5* (Fig. 3c). Furthermore, several epithelial-specific CREs were linked with epithelial marker gene *Shh*, while endothelial-specific CREs were linked with endothelial marker gene *Pecam1* (Fig. 3e). Together, these analyses demonstrate that cell-type-specific genes are tightly regulated by distinct regulatory elements, underscoring their critical role in defining cellular identity during lung development.

Dissecting candidate regulators driving cell fate decision

Lung epithelial and mesenchymal cells are two major cell populations critical for embryonic lung development. To identify lineage-determining regulators, we employed pseudotime trajectory analysis using integrated snRNA-seq and snATAC-seq data[48]. For epithelial cells, Sox9$^+$ lung epithelia localized to the initial stage of the trajectory, while Sox2$^+$ lung epithelia marked the terminal stage. We next inferred transcription regulators positively correlated with the pseudotime trajectory, with TF activity aligning with their expression patterns. This analysis identified a total of 42 TFs regulating epithelial lineage differentiation (Fig. 4a). For example, GATA6, NKX2-1, and SMARCC1 were active in early differentiation stages, whereas SOX2, TRP63, and MEIS1 were active in the later stages.

For mesenchymal cells, we focus specifically on the differentiation of Wnt2$^+$ mesenchymal progenitors into Acta2$^+$ ASM cells. Integrative analysis of gene expression and chromatin accessibility identified 38 TFs associated with ASM differentiation (Fig. 4b). Among these regulators, RUNX1 and RUNX2 showed high enrichment in the late stage. Interestingly, it has been reported that RUNX1 and RUNX2 are active during myofibroblast activation[49,50]. Additionally, the AP-1 complex (comprising FOS and JUN family members) was significant upregulated in both gene expression and TF motif activity during mesenchymal differentiation (Supplementary Fig. 6a). To test AP-1's function in mesenchymal development, we isolated embryonic lung mesenchymal cells and induced smooth muscle cell differentiation by adding TGF-β[51] (Fig. 4c). Analysis of bulk RNA-seq data showed that treatment with the AP-1 inhibitor SR11302 significantly reduced the expression of smooth muscle marker genes, including *Myh11*, *Actg2*, and *Col1a1* (Fig. 4d, e, Supplementary Fig. 6b, and Supplementary Data 9). We also verified their expression through Western blot (Supplementary Fig. 6c). These results suggest that AP-1 acts as an essential regulator of mesenchymal differentiation.

eGRN predicts CTCF as a key regulator of lung morphogenesis

Cellular identity is governed by complex regulatory circuits involving the interplay of genes, TFs, and chromatin dynamics[35]. To decode these circuits, we applied the SCENIC+ algorithm to construct enhancer-driven GRNs (eGRNs), integrating multi-modal data, including chromatin accessibility of CREs, TF binding sites, and transcription levels of target genes[52]. This analysis inferred 142 activator eRegulons (direct: 91; extended: 51), targeting a total of 27,177 regions and 9799 genes (Fig. 5a, Supplementary Fig. 7a, b, and Supplementary Data 10). The GRN revealed important regulators of lineage development, such as NKX2-1, FOXA1, and GRHL2 in Sox9$^+$ lung epithelia; LEF1, FREM1, and FOXF1 in mesenchymal development; and ETV6, SOX7, and SOX17 in endothelial cells (Fig. 5b, and Supplementary Fig. 8)[3,53–55].

Many eRegulons displayed cell-type-specific activity. For instance, NKX2-1 eRegulons were predominantly enriched in lung epithelia, whereas LEF1 and TCF21 eRegulons showed enrichment in mesenchymal progenitors (Supplementary Fig. 7c). CTCF acts as a key chromatin insulator, while RAD21 and SMC3 function as core cohesin components, and they work together to establish and maintain the 3D genome architecture[56]. Analysis of their eRegulons showed broad activity across all cell types (Supplementary Fig. 7c). We further analyzed chromatin accessibility using pseudo-bulk ATAC datasets for epithelial and mesenchymal cells. TF motif analysis revealed that accessible regions in epithelial cells were enriched for lineage-determining TFs like NKX2-1, GRHL2, and KLF6, while mesenchymal cells were enriched for LEF1, TCF4, and MEIS1 motifs (Fig. 5c). Notably, the CTCF motif was strongly enriched in both epithelial and mesenchymal cells. Immunostaining of wild-type E12.5 fetal lung sections confirmed that LEF1 was highly expressed in distal mesenchymal regions, whereas CTCF exhibited widespread expression across all cell types (Fig. 5d). Together, by eGRN inference for lung morphogenesis, we can identify a series of general and cell-type-specific transcription regulators and their target genes.

Genome-wide profiling of CTCF-chromatin binding patterns

To map CTCF-chromatin interactions genome-wide, we performed anti-CTCF CUT&Tag assays using isolated epithelial and mesenchymal cells from E12.5 mouse fetal lungs (Supplementary Fig. 9a). Differential analysis of CTCF peaks identified 6529 epithelial-specific peaks (13.7%

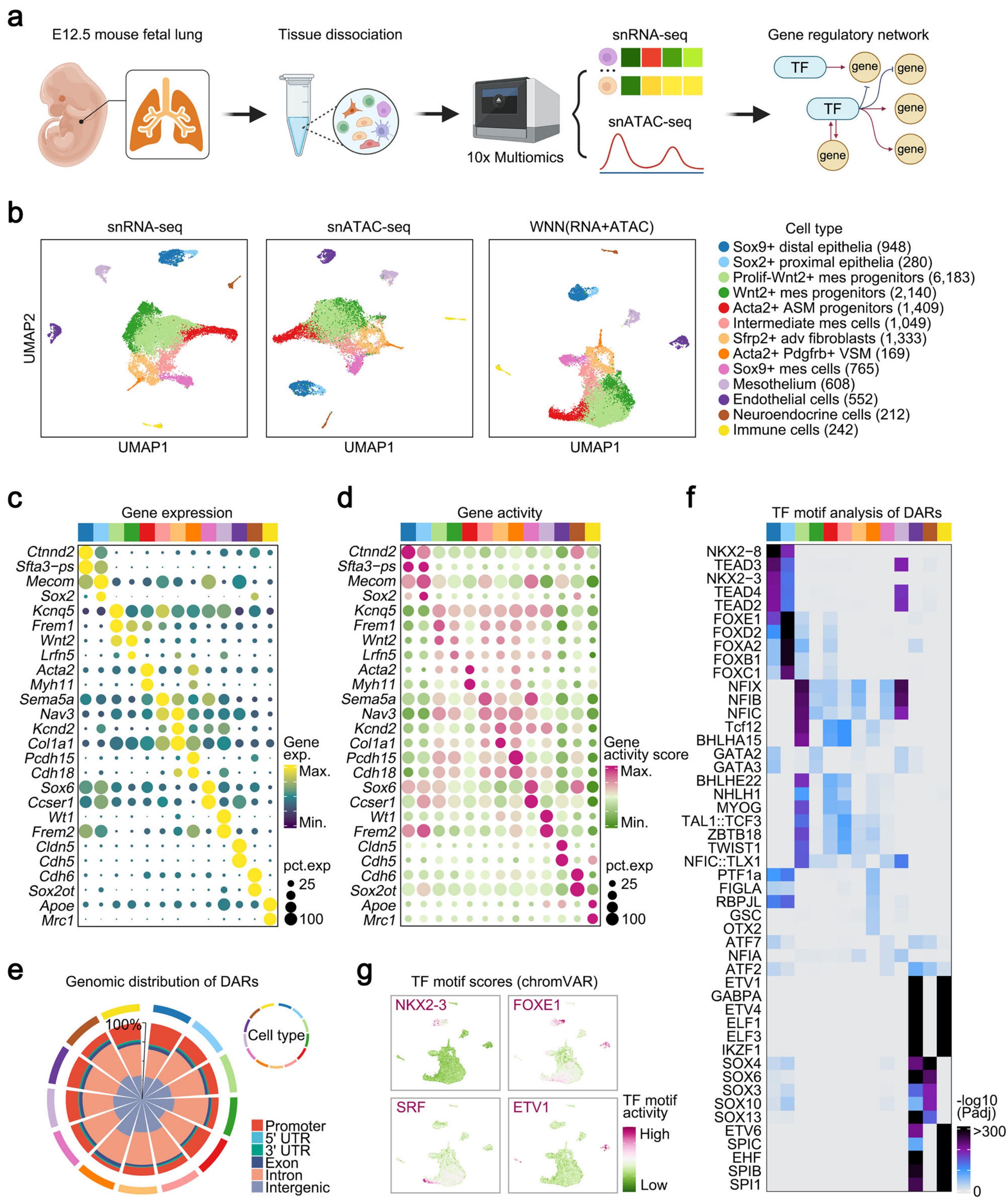


Fig. 2 | Single-nucleus gene expression and chromatin accessibility landscape of mouse lung morphogenesis. a Schematic representation of the experimental workflow; this image was created with BioRender.com. **b** UMAP visualization of snRNA-seq and snATAC-seq, colored by 13 annotated cell types in mouse embryonic lung at E12.5. **c** Dot plot showing snRNA gene expression for each cluster's top 2 marker genes. **d** Dot plot showing matched snATAC gene activity scores for each cluster's top 2 marker genes. **e** Pie plot showing the genomic distribution of differentially accessible chromatin regions (DARs) across various clusters. **f** Heatmap showing the top 5 enriched TF motifs across various clusters. **g** UMAP visualization of TF motif activity of NKX2-3, FOXE1, SRF, and ETV1. Cell types in (**c**–**f**) are labeled using colors as shown in (**b**).

of all epithelial peaks) and 3015 mesenchymal-specific peaks (6.4% of all mesenchymal peaks) (Supplementary Fig. 9b, c). Peak annotation revealed that cell-type-specific CTCF mainly localized in intergenic and intron regions (Fig. 5e). We further analyzed the proportion of CTCF binding on CREs with cell-type-specific versus nonspecific genes expressed in both epithelial and mesenchymal populations (Fig. 5f and Supplementary Fig. 9d). These CTCF-associated marker genes were implicated in cell-type-specific functions. For instance, epithelial

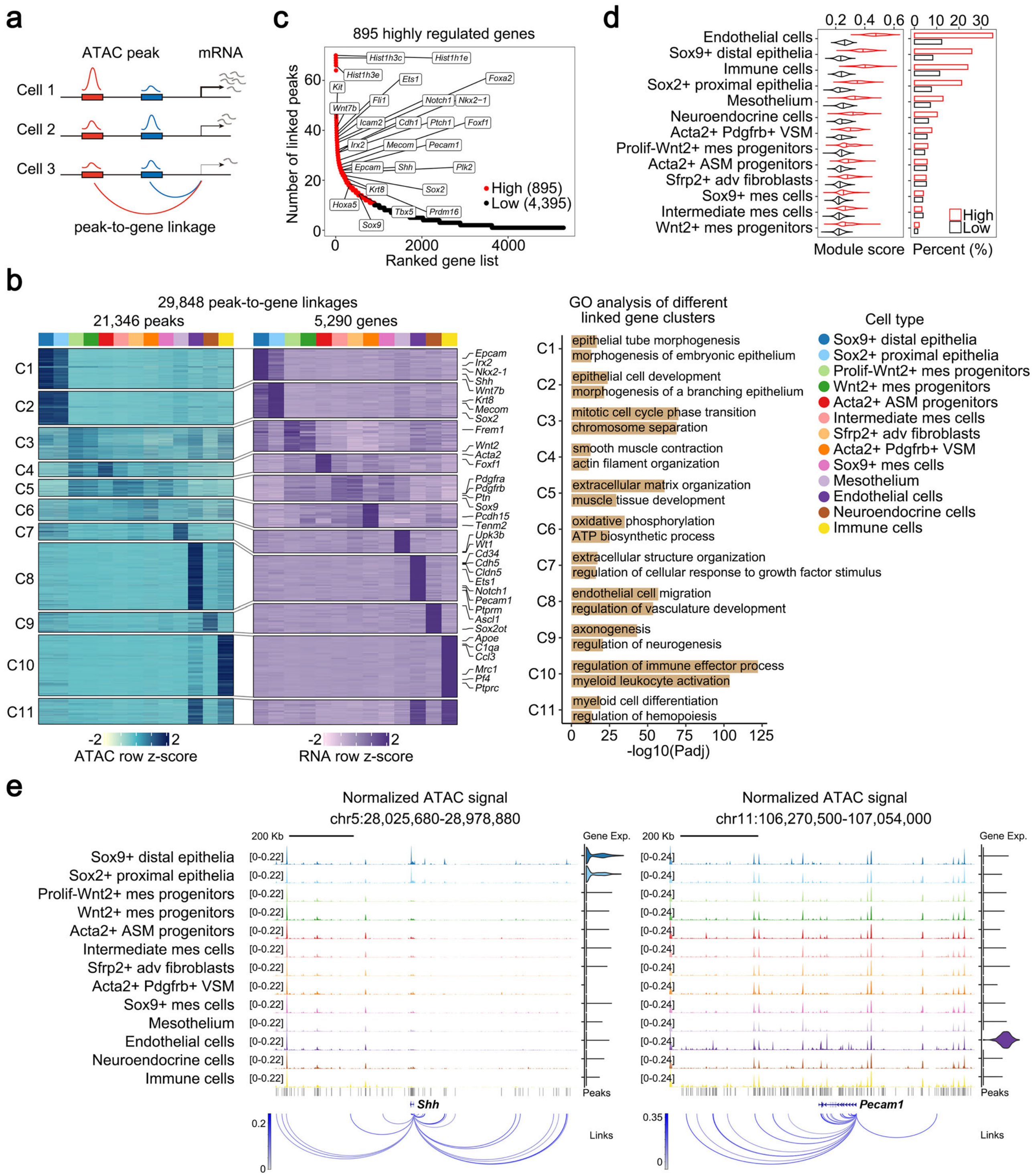


Fig. 3 | Transcriptomic and accessible chromatin signatures across embryonic lung cells. a Schematic representation of the computational method to identify peak-to-gene linkages. **b** Heatmap showing the chromatin accessibility of the linked peaks (left), the RNA expression of the linked genes (middle), and bar plot showing GO term enrichment across different linked gene clusters (right). **c** The classification of linked genes is based on the number of linkages. Genes with more than 10 peaks were designated as highly regulated genes (HRGs; 895 genes), while those with 10 or fewer peaks were classified as lowly regulated genes (LRGs; 4395 genes). Selected cell type marker genes are highlighted. **d** Gene module score analysis of HRGs and LRGs across various cell types (left) and the proportion of HRGs and LRGs in marker genes across various cell types (right). **e** Genome tracks of multi-omics data (gene expression, aggregate ATAC tracks, and peak-to-gene linkages) around the *Shh* and *Pecam1* gene loci (left to right).

marker genes associated with CTCF-binding CREs, including *Shh*, *Wnt7b*, *Fgfr2*, and *Etv5*, were involved in epithelial cell development and branching morphogenesis (Fig. 5g, h). Similarly, mesenchymal marker genes associated with CTCF-binding CREs, including *Acta2*, *Gli1*, *Pdgfrb*, and *Lef1*, were essential for mesenchymal lineage development (Supplementary Fig. 9e, f). These findings demonstrate that, although CTCF is ubiquitously expressed, its chromatin binding patterns exhibit cell-type-specific heterogeneity, suggesting a potential role for CTCF in regulating lineage-specific gene expression and cell fate determination during lung development.

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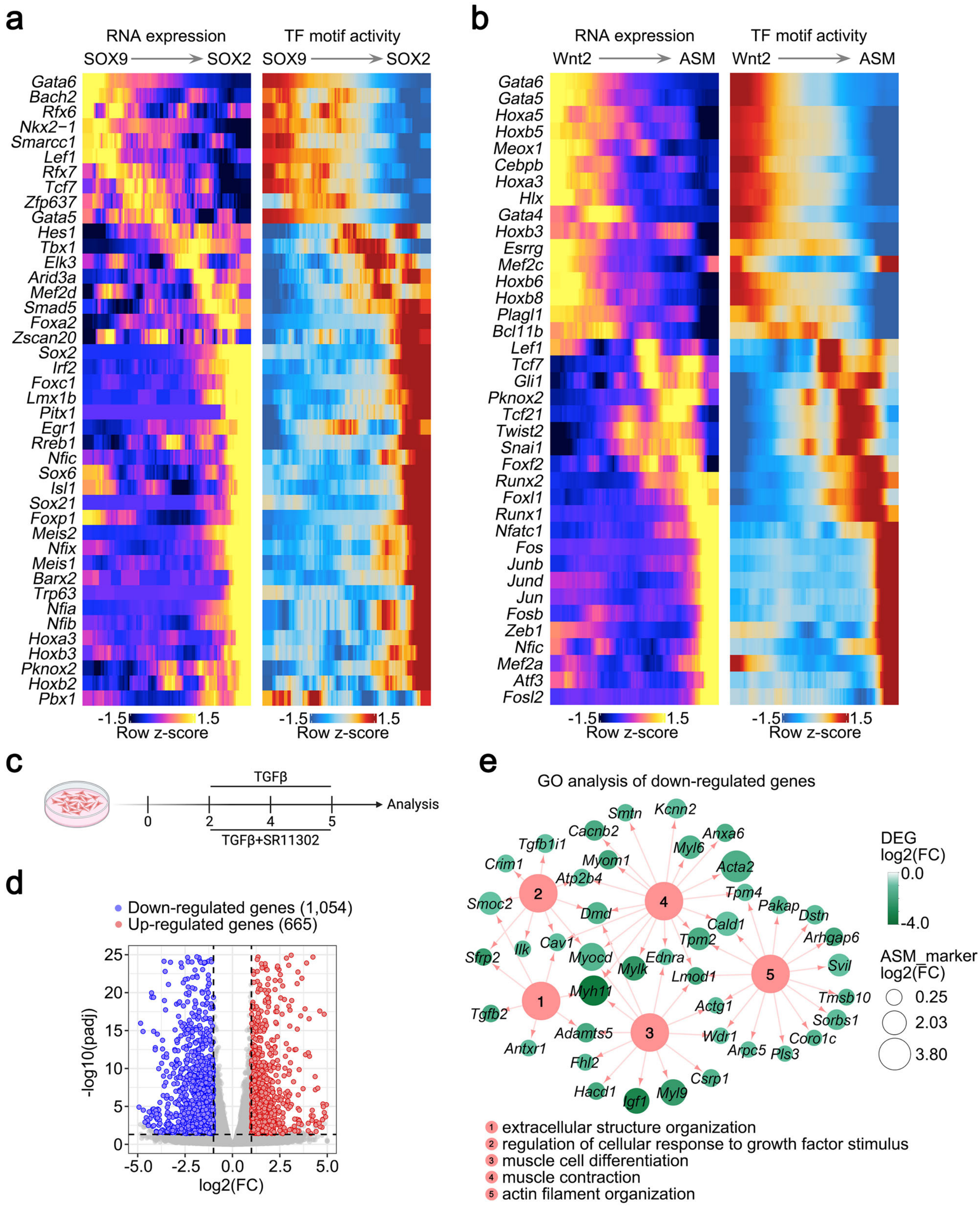

Fig. 4 | Candidate regulators driving embryonic lung epithelial and mesenchymal differentiation. a Paired heatmaps showing matched RNA expression and TF motif activity correlated with Sox9$^+$ distal progenitors to Sox2$^+$ proximal progenitors. **b** Paired heatmaps showing matched RNA expression and TF motif activity correlated with Wnt2$^+$ mesenchymal progenitors to ASM progenitors. **c** Schematic representation of the experimental workflow for in vitro lung mesenchymal cell culture and treatment; this image was created with BioRender.com. **d** Volcano plot showing differentially expressed genes (DEGs) between CTRL and SR11302-treated group ($n = 3$, biological replicates). The significance-adjusted p-value (calculated by DESeq2 using multiple comparisons; Benjamini–Hochberg method, two-sided) is set as <0.05 (horizontal line), and a significant-fold change is defined as >|2| (vertical lines). **e** Functional networks were constructed to visualize the GO enrichment results: nodes represent GO terms and DEGs, whereas edges denote genes' membership in the corresponding GO terms. Node color indicates the log2(FC) of DEGs. Node size indicates the log2(FC) of ASM marker genes.

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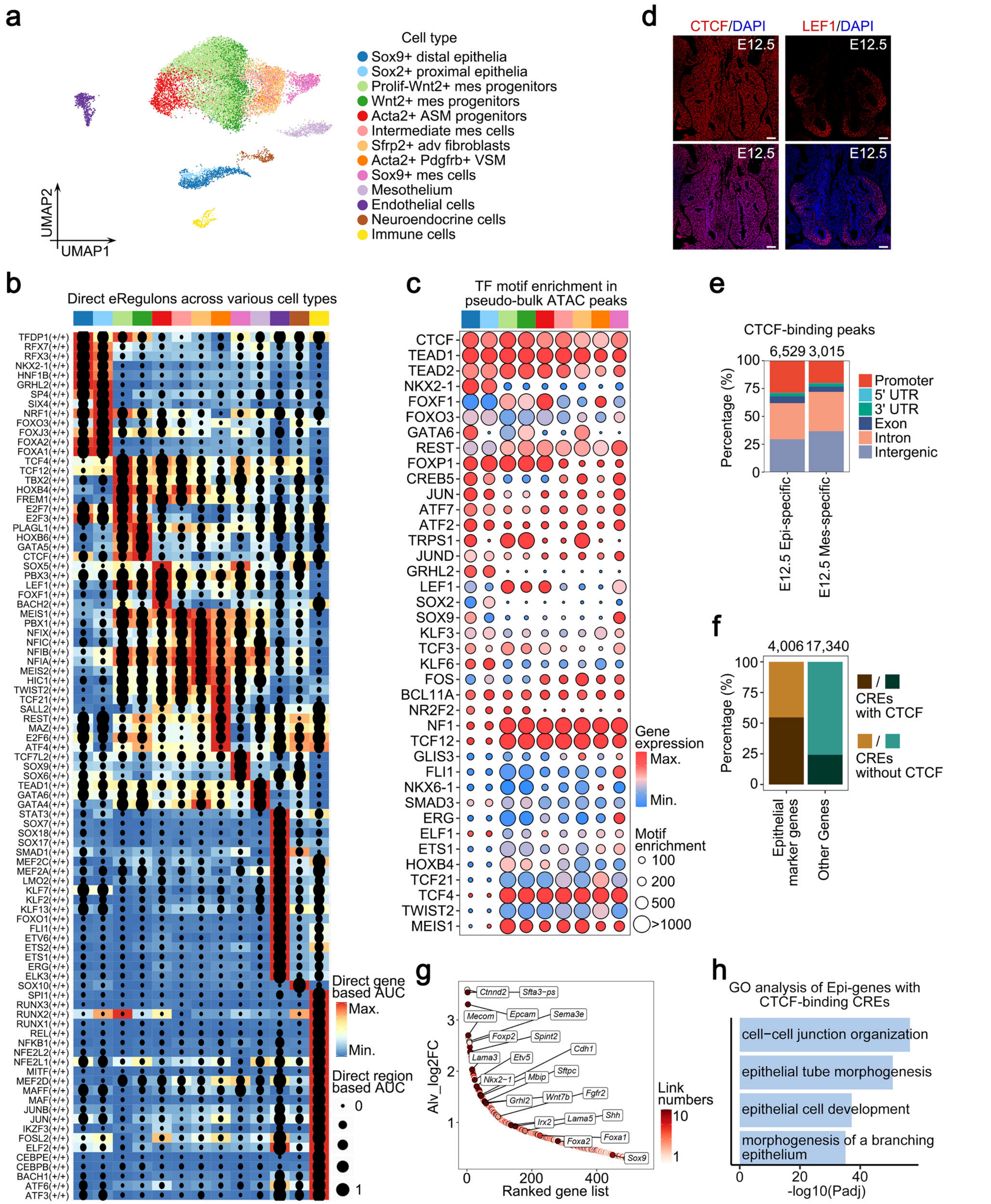


Fig. 5 | Enhancer-driven GRN analysis across embryonic lung cell types. a UMAP visualization of 13 cell types based on eRegulon cell enrichment scores (AUC). **b** Heatmap and dot plot showing direct gene-based AUC of the eRegulon on a color scale and direct region-based AUC of the eRegulon on a size scale. **c** Dot plot showing TF motif enrichment in pseudo-bulk ATAC peaks of epithelial and mesenchymal cells. **d** Immunostaining of CTCF and LEF1 in wild-type mouse lung sections at E12.5 ($n = 3$, biological replicates). Scale bar, 100 µm. **e** Histogram showing the genomic distribution of differential CTCF-chromatin binding peaks. **f** Histogram showing the proportion of CREs with or without CTCF in epithelial marker genes or other genes. **g** Selected epithelial marker genes bound by CTCF are highlighted. **h** Bar plot showing GO term enrichment results of epithelial marker genes with CTCF-binding CREs. Bars, according to the adjusted *p*-value (FDR), reflect the statistical significance of the enrichment. Cell types in (**b**) and (**c**) are labeled using colors as shown in (**a**).

CTCF deletion impairs lung branching and progenitor maintenance

Our data suggested that over half of the epithelial marker genes were potentially regulated by CTCF (Fig. 5f). To investigate the involvement of CTCF in lung epithelial development, we specifically knocked out *Ctcf* in mouse lung endoderm (*Ctcf*-CKO) by use of *Shh-Cre*. Immunostaining results validated high knockout efficiency in both proximal and distal lung epithelial cells (Fig. 6a). *Ctcf*-CKO mice exhibited respiratory abnormalities and died at birth due to respiratory failure. Morphological and histological analyses revealed that mutant lungs were severely hypoplastic, with reduced branching and enlarged airways at E12.5, E13.5, and E14.5 (Fig. 6b and Supplementary Fig. 10a, b). To further assess the function of CTCF in lung branching, we cultured E10.5 fetal lung explants from control and *Ctcf*-CKO mouse embryos. As shown in Fig. 6c, while control explants showed significant growth and branching after 48 h, *Ctcf*-CKO explants failed to branch normally. These results demonstrate that CTCF is essential for lung branching morphogenesis.

We next study the role of CTCF in cell fate specification during lung branching, which relies on proximal-distal patterning[8]. Analysis of SOX9 and SOX2 expression in control and *Ctcf*-CKO lungs at E12.5, E13.5, and E14.5 revealed a progressive decline in Sox2$^+$ progenitors in *Ctcf*-CKO lungs, with few remaining by E14.5 (Fig. 6d). We also observed a reduction in the number of Sox9$^+$ progenitors (Supplementary Fig. 10c). Increased TUNEL-positive epithelial cells, particularly in proximal regions, suggested that *Ctcf* deletion induced apoptosis (Fig. 6e). These findings indicate that CTCF is required for the maintenance of lung progenitors.

To evaluate cell-type-specific gene expression changes upon *Ctcf* deletion, we performed scRNA-seq on whole lung cells from E12.5 control and *Ctcf*-CKO mice. By integrating data from all samples, we ultimately obtained transcriptome information for 11,040 cells, which were annotated into 14 distinct cell types (Fig. 6f, g and Supplementary Fig. 11a–c). Module score analysis revealed suppressed expression of marker genes for Sox9$^+$ progenitor and Sox2$^+$ progenitor in *Ctcf*-CKO lungs (Fig. 6h). Consistent with this, *Ctcf*-CKO lungs exhibited fewer Sox9$^+$ epithelial cells and Sox2$^+$ epithelial cells compared to controls (Fig. 6i).

Interestingly, *Ctcf*-CKO lungs showed an abnormal expansion of *Ebf2*-expressing mesenchymal cells (Supplementary Fig. 12a, b). Cell-cell crosstalk analysis suggested that Ebf2$^+$ mesenchymal cells were regulated by epithelia-secreted WNT7b, which was down-regulated in *Ctcf*-CKO epithelial cells (Supplementary Fig. 12c). Given that *Wnt7b* deletion impaired vascular-associated mesenchymal development[19], reduced *Wnt7b* expression may contribute to abnormal mesenchymal development in *Ctcf*-CKO lungs.

Collectively, these results demonstrate that CTCF plays an important role in lung branching morphogenesis and lung progenitor maintenance by modulating lineage-specific gene expression.

CTCF programs the transcriptome and chromatin accessibility

To elucidate how CTCF regulates lineage-specific gene expression, we isolated lung epithelial cells from E12.5 control and *Ctcf*-CKO mice and performed RNA-seq and ATAC-seq analyses (Fig. 7a). Data from bulk RNA-seq analysis identified 1591 differentially expressed genes (DEGs), encompassing 918 up-regulated and 673 down-regulated genes (Fig. 7b and Supplementary Data 11). Comparative analysis revealed concordance between the overall expression trend of this DEG set in bulk RNA-seq and that in single-cell data: genes upregulated in bulk RNA-seq showed corresponding enrichment in the matching epithelial single-cell clusters, while those downregulated in bulk RNA-seq exhibited consistently reduced expression across both platforms (Fig. 7c and Supplementary Fig. 13a, b). GO term enrichment analysis revealed that up-regulated genes were mainly involved in mesenchymal development, extracellular matrix organization, and apoptosis, while down-regulated genes were associated with epithelial tube morphogenesis and lung development (Fig. 7d).

Signal transduction of FGF10 is central to lung branching morphogenesis[57]. While *Fgf10* expression was slightly upregulated in mesenchymal cells, FGFR2, an essential receptor for FGF10[58], was significantly down-regulated in *Ctcf*-CKO epithelial cells (Fig. 7b and Supplementary Fig. 13c). We verified these results using fluorescent RNAscope (Fig. 7f). Consistently, FGF10 target genes (i.e., *Etv5* and *Anxa4*) were accordingly down-regulated. GO term enrichment analysis further showed the suppression of FGF10 downstream ERK1/2 cascade genes (i.e., *Epha4*, *Dusp10*, and *Atf3*) (Fig. 7e). We also performed phosphorylated ERK (pERK) staining and observed reduced pERK signals in the *Ctcf*-CKO group (Fig. 7g). Further analysis of *Sox2* and *Fgfr2* gene loci revealed that multiple epithelial-specific peak-to-gene linkages were directly bound by CTCF, suggesting that CTCF directly regulates their expression (Supplementary Fig. 14). These findings demonstrate that *Ctcf* knockout disrupts the FGF10-FGFR2-ERK signaling pathway.

Next, we explored how CTCF influences chromatin accessibility. TF footprint analysis confirmed the absence of CTCF footprint signals in *Ctcf*-CKO epithelial cells compared to controls (Fig. 7h). Differential ATAC peak analysis identified 2383 regions with reduced accessibility and 11,707 regions with increased accessibility in *Ctcf*-CKO cells (Fig. 7i). Among these, 98.2% (2342/2383) of regions with reduced accessibility and 65.5% (7672/11,707) of regions with increased accessibility were bound by CTCF (Fig. 7j). Peak annotation revealed that regions with reduced accessibility were predominantly located in regulatory elements (introns and intergenic regions) rather than promoters (Fig. 7k). We then assessed the physical distance between these differentially accessible regions and their closest CTCF binding peaks. Our findings showed that regions with decreased accessibility were positioned closer to the nearest CTCF binding peaks (median distance ~0 bp), indicating these are direct CTCF-bound loci (Fig. 7l). By comparison, regions with increased accessibility, while farther from CTCF peaks than the reduced-accessibility regions, were significantly closer to these peaks than unchanged regions. We also performed motif analysis to study which TFs bind these differentially accessible regions. These results showed that CTCF and BCL11A binding motifs were enriched in regions with reduced accessibility, while TEAD and FOX family TF motifs were enriched in regions with increased accessibility (Fig. 7m). BCL11A has been reported to function as a 3D genome regulator, while YAP/TEAD and FOXA1/FOXA2 are essential for lung development and cell specification[26,59,60]. These results further support the notion that CTCF plays an essential role in both chromatin organization and lung morphogenesis.

Collectively, these findings highlight the critical role of CTCF in programming gene expression and chromatin accessibility, which is essential for lung epithelial lineage determination (Supplementary Fig. 15).

Discussion

Cell-type-specific gene expression and chromatin dynamics are crucial for determining cell fate and organogenesis. However, the mechanisms by which chromatin dynamics regulate lineage-specific transcriptional programs during lung morphogenesis remain poorly understood. In this study, we employed paired single-cell multi-omics technologies to simultaneously profile gene expression and chromatin accessibility in mouse lung development. We inferred gene regulatory programs dictating lung morphogenesis and identified key regulators of epithelial and mesenchymal differentiation. Using a genetic knockout mouse model, we provided compelling evidence for the pivotal role of CTCF in regulating lung branching morphogenesis and lung progenitor maintenance. Our study provides a valuable resource for further investigation of lung morphogenesis and highlights how specific chromatin factors orchestrate cell fate and organ development.

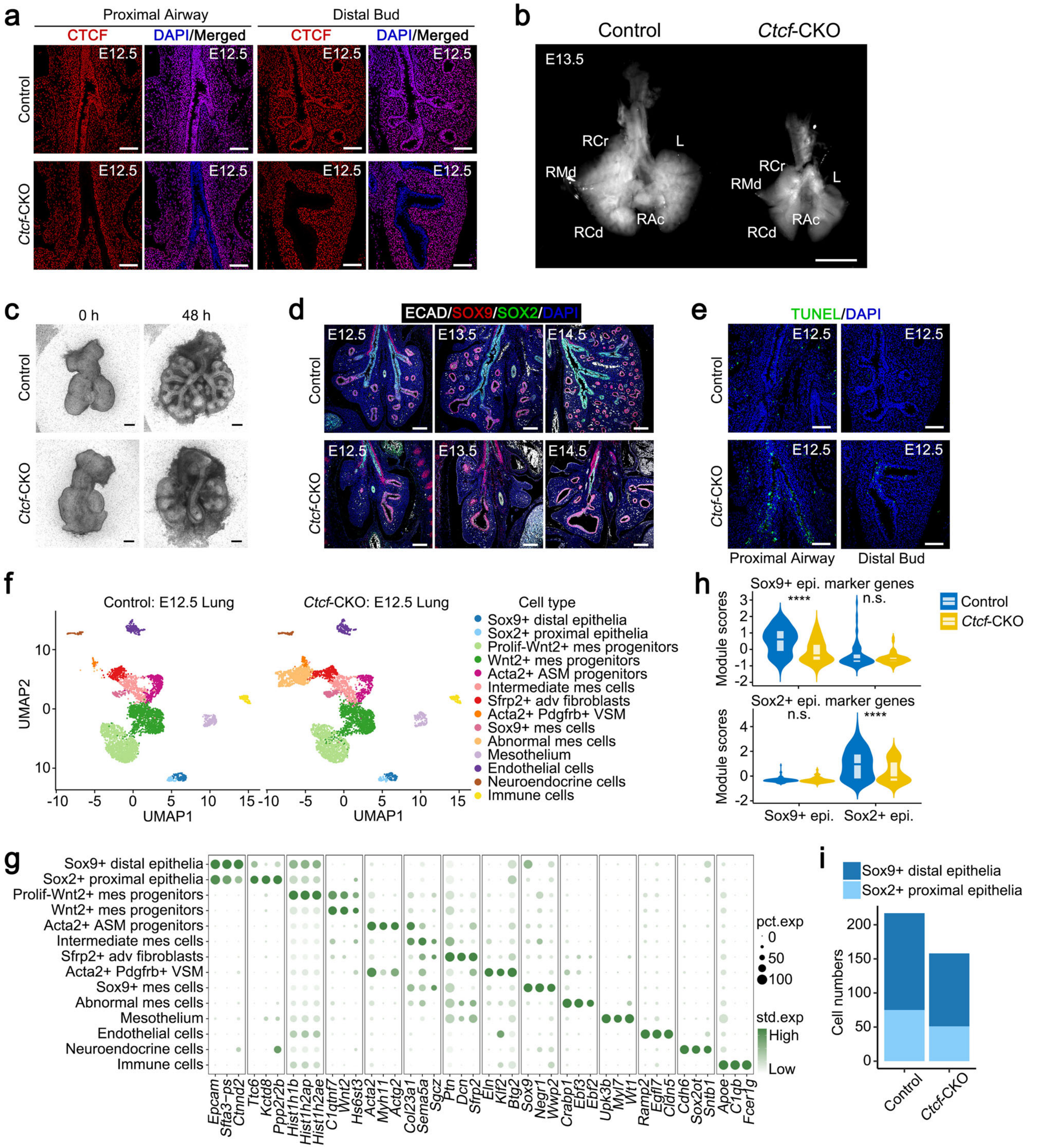
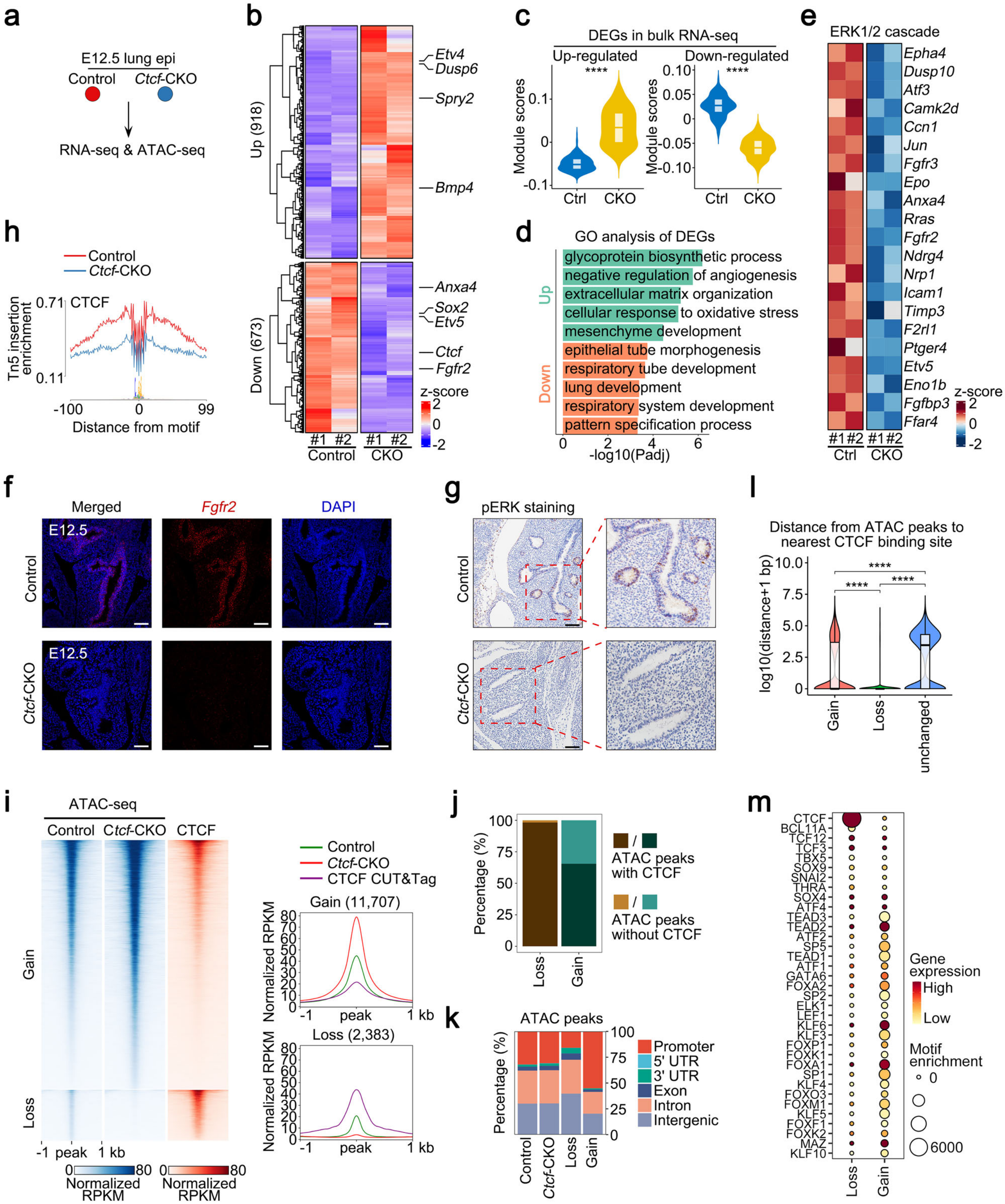


Fig. 6 | CTCF controls lung branching morphogenesis and progenitor maintenance. a Immunostaining of CTCF in lung sections from control and *Ctcf*-CKO mice at E12.5 ($n = 3$, biological replicates). Scale bar, 100 μm. **b** Bright-field imaging of control and *Ctcf*-CKO lungs at E13.5 ($n = 3$, biological replicates). RCr right cranial, RMd right middle, RAc right accessory, RCd right caudal, L left. Scale bar, 500 μm. **c** Bright-field imaging of control and *Ctcf*-CKO lung explants cultured for 48 h ($n = 3$, biological replicates). Scale bar, 200 μm. **d** Immunostaining of E-Cadherin (ECAD), SOX9, and SOX2 in control and *Ctcf*-CKO lung sections at E12.5, E13.5, and E14.5 ($n = 3$, biological replicates). Scale bar, 200 μm. **e** TUNEL staining in control and *Ctcf*-CKO lung sections at E12.5 ($n = 3$, biological replicates). Scale bar, 100 μm. **f** UMAP visualization of the annotated cell types captured from control and *Ctcf*-CKO lungs at E12.5. **g** Dot plot showing the expression of marker genes across various cell clusters. **h** Violin plots showing module scores of $Sox9^+$ progenitors marker genes set and $Sox2^+$ progenitors marker genes set from control and *Ctcf*-CKO lungs at E12.5. Statistical analysis was performed using two-sided Student's *t* tests. The *p*-values are 5.3e−11 and 0.055, respectively (top); the *p*-values are 0.89655 and 0.00037, respectively (bottom). **i** Bar plot showing the numbers of $Sox9^+$ progenitors and $Sox2^+$ progenitors in control and *Ctcf*-CKO lungs at E12.5.

Proximal-distal patterning is fundamental to lung branching morphogenesis[3], and the maintenance of lung progenitors is critical to this process. It has been shown that disruption of epithelial progenitor maintenance would lead to branching defects[9,33]. While previous studies have elucidated transcriptional dynamics during lung progenitor development[39,53], our work highlights that the lineage specification relies on active maintenance of progenitor identity, mediated through extensive chromatin reorganization involving changes in accessibility,



peak-to-gene linkages, and regulatory factor associations. These observations underscore the critical role of chromatin regulators in preserving progenitor pools during lung morphogenesis. While it has recently been reported that a CTCF mutation at R567 results in lung developmental defects characterized by non-inflated alveoli[61], the functions of CTCF in early lung morphogenesis remain unknown. In this study, we found that epithelial-specific deletion of *Ctcf* disrupts the maintenance of lung progenitors. Multi-omics analyses revealed that CTCF is required for sustaining the expression of marker genes defining both Sox9$^+$ progenitors and Sox2$^+$ progenitors. CTCF directly binds promoters and intergenic regions of these genes, and its loss impairs the chromatin accessibility necessary to maintain their expression. Notably, loss of the miRNA regulator DICER also compromises progenitor maintenance and lung branching morphogenesis[27], suggesting that coordinated epigenetic mechanisms collaborate to modulate lineage-specific gene activation and repression during progenitor maintenance. Together, these findings provide mechanistic insights into how chromatin regulation safeguards lung progenitor maintenance and lung morphogenesis.

CTCF is a key regulator of enhancer-promoter interactions and the organization of the 3D genome[56,62–65]. Recent studies have

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Fig. 7 | CTCF is essential for programming the transcriptome and chromatin dynamics. a Schematic representation of the experimental workflow. **b** Heatmap showing differentially expressed genes (DEGs) of lung epithelial cells in bulk RNA-seq following *Ctcf* deletion at E12.5. **c** Violin plots showing module scores of bulk RNA-seq DEGs set in Sox9^{+} progenitors and Sox2^{+} progenitors from control and *Ctcf*-CKO lungs at E12.5. Ctrl = 217 and CKO = 158 biologically independent cells were analyzed. In the left half of the graph, for the Ctrl group, the distribution was: min. = −0.08, 25th percentile = −0.06, median = −0.05, 75th percentile = −0.04, and max. = 0.02; and for the CKO group: min. = −0.04, 25th percentile = 0.00, median = 0.03, 75th percentile = 0.07, and max. = 0.13. *p*-value = 7.45e−61 in Student's *t* tests. In the right half of the graph, for the Ctrl group, the distribution was: min. = −0.03, 25th percentile = 0.01, median = 0.03, 75th percentile = 0.04, and max. = 0.08; and for the CKO group: min. = −0.11, 25th percentile = −0.07, median = −0.06, 75th percentile = −0.04, and max. = −0.01. *p*-value = 1.84e−128 in Student's *t* tests. **d** Bar plot showing GO term enrichment results of up-regulated genes and down-regulated genes. **e** Heatmap showing the expression of ERK1/2 cascade genes in lung epithelial cells from control and *Ctcf*-CKO lungs. **f** *Fgfr2* in situ hybridization in lung sections from control and *Ctcf*-CKO mice at E12.5 ($n = 3$, biological replicates). Scale bar, 100 μm. **g** Immunostaining of pERK in lung sections from control and *Ctcf*-CKO mice at E12.5 ($n = 3$, biological replicates). Scale bar, 100 μm. **h** Visualization of CTCF footprint in accessible chromatin of lung epithelial cells from control and *Ctcf*-CKO lungs. **i** Heatmaps and mean plots showing differential ATAC peaks and CTCF-binding signals. **j** Histogram showing the proportion of differential ATAC peaks with or without CTCF. **k** Histogram showing the genomic distribution of ATAC peaks in different groups. **l** Violin plots showing the distance from ATAC peaks to nearest CTCF binding sites. 2000 peaks were randomly selected from each of the Gain, Loss, and unchanged peak sets. Statistical significance was assessed using two-sided Wilcoxon rank-sum test (Mann–Whitney U test). When Gain is compared with Loss, the *p*-value is 6.42e−205. When Gain is compared with unchanged, the *p*-value is 3.60e−49. When Loss is compared with unchanged, the *p*-value is ~0. **m** Dot plot showing TF motif enrichment in differential ATAC peaks between control and *Ctcf*-CKO epithelial cells.

highlighted its context-dependent roles in various developmental processes, suggesting lineage-specific functions[66–75]. Therefore, the mechanisms by which CTCF regulates lineage-specific gene expression, cell fate decisions, and organogenesis attract intensive studies. Here, we integrated single-cell multi-omics with CUT&Tag to investigate CTCF's role in embryonic lung development. We found that CTCF directly and selectively binds gene regulatory elements to precisely control the expression of lung endoderm-essential genes, including *Fgfr2*, *Etv5*, *Anax4*, and *Shh*. Importantly, CTCF loss disrupts the FGF10-FGFR2-ERK signaling cascade, a crucial pathway in lung morphogenesis, further underscoring the central role of CTCF in regulating lung endoderm-specific gene expression and lung development.

Promoter-enhancer (P-E) interactions mediated by CTCF vary across cell types[76,77], though the underlying mechanisms are intricate and largely unclear. We observed heterogeneity in CTCF-chromatin binding patterns across different cell types. Intriguingly, P-E interactions and regulatory elements for the same gene, such as *Sox2*, differ between Sox2^{+} lung epithelia and neuroendocrine cells, highlighting dynamic changes in P-E interactions across cell types and developmental stages. We also revealed that CTCF is essential for the regulation of chromatin accessibility, consistent with previous reports that CTCF depletion in human B-ALL cell line disrupts chromatin accessibility[78]. Interestingly, many increased accessible regions were not directly bound by CTCF but closer to CTCF-binding sites, likely due to the loss of long-range interactions or domain boundaries maintained by CTCF, which secondarily affects accessibility at distal loci. Future studies could explore how CTCF interacts with other regulatory factors and modulates cell-cell interactions to shape lung development. Additionally, integrating single-cell chromatin conformation capture (scHi-C) techniques could provide further insights into the regulatory functions of CTCF in lineage-specific gene expression.

In summary, this study utilizes single-cell multi-omics techniques to dissect key gene regulatory programs in lung development at single-cell resolution. Leveraging this multi-modal dataset, we identified and predicted a series of transcription regulators that may be involved in lung development. We further demonstrated that CTCF is required for the maintenance of lung progenitors and lung branching morphogenesis by modulating gene expression and chromatin accessibility. These findings provide critical mechanistic insights into how chromatin dynamics govern cell fates and organogenesis.

Methods

Mouse strains

Ctcf$^{flox/+}$ (C57BL/6) mice were generated by GemPharmatech Co., Ltd. (Strain No. T016636); *Shh-Cre* (C57BL/6) mice were purchased from the Jackson Laboratory (Strain No. 005622). Mice were mated in the IPM-GBA animal facility to generate *Ctcf*$^{flox/flox}$;*Shh*$^{Cre/+}$ pups and littermate controls. In this study, mice at 2–6 months old were used for time matings. Observance of a vaginal plug was designated as E0.5. All mice were housed under specific pathogen-free (SPF) conditions with a 12/12-hour light/dark cycle, a constant temperature of 22 °C, and a relative humidity of 50%. All experimental procedures were approved by the Animal Care and Use Committee of the Greater Bay Area Institute of Precision Medicine (Ethics approval number: IPMGBA-24-0024). Genotyping primers used in this study were listed in Supplementary Table 1.

Single-cell suspension preparation

Timed-pregnant mice were sacrificed at the indicated time points. The uterus was removed, and embryos were harvested on ice. The embryonic lung was incubated in digestion medium [DMEM (Hyclone, SH30243.01) + 100 U/mL DNase I (NEB, M0303S) + 100 μg/mL Liberase™ (Roche, 5401119001)] for 20 min at 37 °C at 300 rpm. After digestion, lung tissue was dissociated into single cells via pipetting. The cells were washed and resuspended in 1 × PBS + 2% FBS (Hyclone, SH30096.03).

Embryonic lung cell sorting

Lung cell suspension (see above) was incubated with antibodies on ice for 20–25 min in 1×PBS with 2% FBS. Before cell sorting, the suspension was filtered through 40 μm cell strainers. Cells were sorted on a FACS BD Aria III flow cytometer. Epithelial cells were defined as lived CD45^{-}CD31^{-}CD326^{+} cells and mesenchymal cells were defined as lived CD45^{-}CD31^{-}CD326^{-} cells. FACS data were analyzed using FlowJo software (Tree Star). Antibodies used in this study were listed in Supplementary Table 3.

Culture and treatment of embryonic lung mesenchymal cells

Lung cell suspension (see above) was washed with 1×PBS and seeded in 24-well plates with 100,000 cells per well. Mesenchymal cells were cultured in DMEM/F12 (Gibco, 12634010) supplemented with Penicillin/Streptomycin (Gibco, 15140122), 1×B27 supplement (Gibco, 17504044), 1×N2 supplement (Gibco, 17502048) and 1.25 mM N-acetylcysteine (Sigma-Aldrich, A0737). Two days later, the medium was changed and 10 ng/mL TGF-β (PeproTech, 100-21C) was added. After another 3 days, cells were harvested and analyzed.

Culture of embryonic lung explant

Timed-pregnant mice were sacrificed at E10.5. The embryonic lung was isolated as above. 24-well plates were prepared with 400 μL DMEM/F12 + 1% P/S per well. A Nuclepore Polycarbonate Track-Etch Membrane (Whatman, 10417501) was placed on top of the medium (the shiny side down, rough side upwards). Embryonic lungs were placed on top of the Nuclepore Polycarbonate Track-Etch Membrane. The lung explant was imaged by Leica DM IL LED Fluo microscope.

Histological and immunofluorescent analysis

Embryos were harvested at E12.5, E13.5, and E14.5 and fixed whole in 4% PFA overnight on a rocker at 4 °C. Whole embryos were washed with 1×PBS three times for 10 min, then dehydrated, embedded in paraffin and sectioned at a thickness of 5 µm. The sections were rehydrated, antigen retrieved and blocked with 5% BSA (Sigma-Aldrich, B2064) in PBS. Finally, they were incubated overnight with primary antibodies in 1×PBS + 0.3% Triton X-100 (Sigma-Aldrich, T9284) + 1% BSA at 4 °C. The sections were washed and then incubated at room temperature with appropriate secondary antibodies and DAPI (Sigma-Aldrich, D9542) for 1 h. After washing, sections were mounted with mounting medium Fluoromount-G (Southern Biotech, 0100-01) and imaged with a Leica Stellaris 5 confocal microscope.

For immunohistochemical staining, sections were rehydrated, subjected to antigen retrieval, and treated with 3% hydrogen peroxide (H_2O_2) for 10 min to quench endogenous peroxidase activity. Subsequent procedures were carried out in accordance with the manufacturer's instructions of the immunohistochemistry detection kit (ABclonal, RK05872). Imaging was performed using an ECLIPSE Ni-U microscope (Nikon, Japan). Antibodies employed in this study were listed in Supplementary Table 3.

RNAscope analysis

RNAscope in situ hybridization was conducted on E12.5 tissue sections using a probe targeting *Fgfr2* (ACD, 443501-T1), following the manufacturer's protocol for the RNAscope® Multiplex Fluorescent Reagent Kit v2 (ACD, 323100). Briefly, sections were deparaffinized and then treated with hydrogen peroxide for 10 min at room temperature, followed by heat-induced target retrieval (30 min in boiling buffer) and protease digestion (Protease Plus, 30 min at 40 °C). Subsequently, sections were hybridized with the target probe in a HybEZ™ oven (ACD) for 2 h at 40 °C. Signal amplification was performed using the kit reagents. Finally, sections were counterstained with DAPI for 30 s at room temperature and mounted with Fluoromount-G mounting medium (Southern Biotech, 0100-01). Images were acquired using a Leica Stellaris 5 confocal microscope.

Quantitative RT-PCR analysis

Total RNA was extracted using the RNeasy Mini-plus Kit (QIAGEN, 74134) according to the manufacturer's instructions. Reverse transcription was conducted using GoScript Reverse Transcription System (Promega, A5001). Quantitative RT-PCR was performed using 2×SYBR Green qPCR Master Mix (Selleck, B21202). Relative mRNA levels were normalized to the Gapdh mRNA levels. qPCR primers used in this study were listed in Supplementary Table 2.

Single-cell multi-omics library and sequencing

Single-cell multi-omics libraries were constructed using Chromium Next GEM Single Cell Multiome ATAC + Gene Expression Reagent Kits (10x Genomics, PN-1000283) for 2 independent biological replicates according to the manufacturer's protocol. Briefly, after dissociation, cells (sorted epithelial cells: whole lung cells = 1:10) were collected and resuspended in a chilled Lysis Buffer to isolate nuclei. After washing, nuclei suspensions underwent incubation in a Transposition Mix containing transposase. The transposase penetrated the nuclei and selectively cleaved DNA within the open chromatin regions. Simultaneously, adapter sequences were appended to the ends of the DNA fragments. The Single Cell Multiome ATAC + GEX Gel Beads include a poly(dT) sequence that enables production of barcoded, full-length cDNA from poly-adenylated mRNA for gene expression library and a Spacer sequence that enables barcode attachment to transposed DNA fragments for ATAC library. Finally, the snATAC-seq libraries were sequenced on the Element AVITI instrument, and the snRNA-seq libraries were sequenced on the NovaSeq 6000 platform by NovelBio.

CUT&Tag library construction and sequencing

CUT&Tag libraries were generated using the CUT&Tag Assay Kit (ABclonal, RK20265) following the manufacturer's guidelines. In brief, cells were captured by Concanavalin A Magnetic Beads, then incubated with primary antibodies at 4 °C, and then bound to pAG-Tn5. The fragmented DNA was purified and amplified through PCR to form the libraries. Prepared libraries were sequenced on the Illumina NovaSeq 6000 platform with paired ends and 150 bp read lengths. Each sample includes two independent biological replicates, and each replicate library produced at least 40 M reads for comprehensive bioinformatics analysis.

RNA-seq library construction and sequencing

Total RNA from isolated epithelial cells was extracted using RNeasy Plus Mini Kit (QIAGEN, 74134). The fresh RNA was purified into mRNA using VAHTS mRNA Capture Beads (Vazyme, N401). Then, the captured mRNA underwent library generation using Equalbit RNA HS Assay Kit (Vazyme, NR605) following the manufacturer's instructions. Prepared RNA-seq libraries were sequenced on the Illumina NovaSeq 6000 platform. Each sample includes two independent biological replicates, and each replicate library produced at least 40 M reads for comprehensive bioinformatics analysis.

ATAC-seq library construction and sequencing

Epithelial nuclei were isolated and fragmented with Tn5 transposase. DNA was isolated using the DNA Clean and Concentrator kit (Zymo, D4013) and subjected to library construction using Q5 high-fidelity DNA polymerase (New England Biolabs, M0544S). For high-throughput sequencing, over 40 million reads were obtained per sample using the Illumina NovaSeq 6000 platform for 2 independent biological replicates.

scRNA-seq library construction and sequencing

Single-cell transcriptomic libraries were constructed using Chromium Single Cell 3' Reagent Kits v3 (10× Genomics) according to the manufacturer's protocol. Briefly, after dissociation, whole lung cells were collected and resuspended to form a cell suspension, and the cell concentration was adjusted to 1000 cells/µL (viability ≥85%). Approximately 10,000 cells were captured in droplets and then lysed to release total RNA. RNA was barcoded through reverse transcription from individual gel beads in emulsion. The cDNA was then amplified for the library construction. Prepared libraries were sequenced on the Illumina NovaSeq 6000 by Berry Genomics. Each library produced at least 100 GB of raw data for downstream bioinformatics analysis.

CUT&Tag data analysis

For analyzing CUT&Tag data, we used the ENCODE ChIP-seq pipeline (v2.2.1, https://github.com/ENCODE-DCC/chip-seq-pipeline2). The raw sequencing data were processed to remove adapters and low-quality sequences using Cutadapt (v2.5) with the parameters: -m 5 -e 0.1. Then, clean reads were aligned against the mouse mm10 genome using bwa (v0.7.17). Then, sam files were converted to bam files using samtools (v1.9). PCR duplicates were removed using Picard (v2.20.7, https://broadinstitute.github.io/picard/). Peaks were called using MACS2 (v2.2.4). The conservative narrow peaks were used for downstream analysis. For visualization, bam files were converted to bigWig files using deepTools (v3.3.1) under 50 bp resolution. Besides, RPKM was used to normalize the number of reads per bin. We used the R package DESeq2 (v1.38.3) for differential peak analysis. Regions with p-value < 0.05 and fold-change ≥1.5 were considered as differentially binding peaks. We used R package ChIPseeker (v1.34.1) for peak annotation. We used ComputeMatrix and plotHeatmap of deepTools to generate count matrices and heatmaps, respectively. TF motif enrichment was calculated using HOMER (v4.11). For the correlation between replicates of each experiment, we used deepTools to calculate Spearman's correlation.

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RNA-seq data analysis
We performed RNA-seq analysis as described previously[79]. Briefly, after removing adapters using Cutadapt (v2.5), paired-end reads were aligned to the mm10 mouse genome using Hisat2 (v2.2.1) in paired-end mode with default parameters. Expression levels of genes were calculated using StringTie (v2.2.1). Subsequently, raw read counts for genes were calculated using the Python script prepDE.py3. Differentially expressed genes were identified using the R package DESeq2 (v1.38.3). Genes with read counts >50 in at least one sample were kept for further analysis. A given gene was considered to have significantly changed if the adjusted P value (padj) is <0.05, the P value is <0.01, and the fold-change is ≥1.5.

Gene Ontology (GO) enrichment analysis
All Gene Ontology (GO) term analyses in this study were systematically conducted using the R package clusterProfiler (v4.6.2), with rigorous statistical testing integrated throughout the workflow to ensure the reliability of enrichment results. Briefly, the analysis steps were as follows: first, gene symbols (in SYMBOL format) were converted to ENTREZ IDs using the 'bitr' function of clusterProfiler to ensure compatibility with subsequent enrichment analyses. Next, GO term enrichment analysis was performed using the 'enrichGO' function, where key parameters were set as follows: the organism-specific annotation database was specified as 'org.Mm.eg.db'; the Benjamini–Hochberg (BH) method was applied for multiple-testing correction to control the false discovery rate (FDR); and the statistical significance of enriched GO terms was strictly defined by a corrected P-value (padj) threshold of <0.01. Only GO terms meeting this padj criterion were considered as significantly enriched functional categories for the input gene sets, ensuring that the reported GO term associations were not driven by random chance.

ATAC-seq data analysis
ATAC-seq data were analyzed using a standardized ENCODE ATAC-seq pipeline (v2.2.2, https://github.com/ENCODE-DCC/atac-seq-pipeline). The mouse reference genome (mm10) was used in the pipeline. Trimming, mapping, and duplicate-removing were performed as the ENCODE pipeline suggested using Cutadapt (v2.5), Bowtie2 (v2.3.4.3), and Picard (v2.20.7, https://broadinstitute.github.io/picard/), respectively. MACS2 (v2.2.4) was used for peak calling with parameters: --nomodel --shift 100 --extsize 200 --pval 0.01 --SPMR --call-summits. Differential peak analysis was performed by R package DiffBind (v3.8.4, http://bioconductor.org/packages/release/bioc/html/DiffBind.html). Then, regions with BH-adjusted P value (FDR) < 0.05 and FC > 0 were considered as differentially accessible regions (DARs) in the subsequent analysis.

We used the footprinting command of rgt hint (v1.0.2) to perform transcription factor footprint analysis on ATAC-seq data with main parameters: --atac-seq --paired-end --organism=mm10.

scRNA-seq data analysis
Dimensionality reduction and cell annotation. FASTQ files were run through CellRanger (v7.1.0) software with default parameters for demultiplexing, aligning reads with STAR software to mm10, and counting unique molecular identifiers (UMIs). As input files of the Seurat R package (v4.4.0), the filtered gene expression matrices were then used for downstream analyses. Low-quality cells were filtered (expressing <600 or >8000 unique gene counts, and >5% mitochondrial reads). Raw RNA counts were normalized and scaled via SCTransform, with percentage of mitochondrial reads and nCount_RNA regressed out. Subsequent analyses, including PCA, sample-based CCA batch integration, nearest neighbor clustering, and UMAP embedding construction, were conducted in Seurat. Cell types were manually assigned by merging raw clusters based on differential expression and knowledge of canonical gene markers. The 'FindAllMarkers' function was used to calculate cluster-specific genes (|avg_logFC|> 0.25 and p_val_adj <0.05). The 'AddModuleScore' function was used to calculate gene set module scores in single cells with default parameters.

Differential expression analysis of single-cell data was performed using the Wilcoxon rank-sum test (Mann–Whitney U test), a non-parametric method appropriate for comparing gene expression distributions across distinct cell populations.

Pseudotime trajectory analysis
We inferred the developmental timeline of the E12.5 fetal lung mesenchyme. First, the Seurat object was converted to an AnnData object compatible with Scanpy (v1.8.2), and pseudotime analysis was conducted in a Python environment. Specifically, we used Scanpy to generate a diffusion map and calculate diffusion pseudotime (DPT). While diffusion map analysis already revealed three distinct differentiation trajectories within the mesenchyme, we further validated the accuracy of these pseudotime results by re-analyzing mesenchymal cell trajectories using Palantir (v1.4.1)[80]. For the Palantir workflow, a normalized count matrix was exported and processed in Python; Prolif-Wnt2$^+$ mes progenitors were designated as the starting cell state, and no terminal states were pre-specified. Pseudotime values, entropy, and diffusion map embedding coordinates were subsequently plotted for visualization. Additionally, MAGIC was employed for single-cell data imputation to improve the clarity of visualizations.

Cell-cell communication analysis
We performed cell-cell communication analysis using the R package CellChat (v2.1.1). Firstly, we created a CellChat object by utilizing the 'createCellChat' function. Subsequently, we calculated the communication probabilities using the 'computeCommunProb' function and filtered out cell-cell interactions detected in fewer than 10 cells. To investigate cell-cell interactions at the signaling pathway level, we employed the 'computeCommunProbPathway' function, ultimately obtaining the ligand-receptor information for the intercellular interactions.

Single-cell multi-omics (snRNA-seq+snATAC-seq) data analysis
Quality control, dimensionality reduction and identification of fetal lung cell types. Raw fastq files were aligned to the mouse mm10 reference genome using CellRanger ARC (v2.0.2). ATAC fragments were processed with Signac (v1.13.0), while RNA data were processed using Seurat (v4.4.0). Quality control filtering retained nuclei meeting the following criteria: TSS enrichment > 4, number of fragments > 1000, 500 < nFeature_RNA < 8000, 1000 < nCount_RNA < 40,000, and mitochondrial gene percentage <15%. Post-quality control nuclei underwent doublet removal using DoubletFinder (v2.0.4). Raw RNA counts were normalized and scaled with SCTransform, incorporating regression of mitochondrial read percentage and nCount_RNA. Subsequent analyses in Seurat included PCA, CCA-based batch integration (stratified by sample), nearest neighbor clustering, and UMAP embedding construction. Cell types were annotated using differential expression and knowledge of canonical gene markers. The RNA-derived annotations, normalized gene expression matrix, and UMAP embeddings were added to the Signac project onto the nuclei with the same barcodes.

Following the Signac documentation, after constructing a unified peak set, we normalized the single-nucleus ATAC-seq data using the 'RunTFIDF' function. Subsequently, we employed the 'FindTopFeatures' function to calculate highly variable features and used the 'RunSVD' function for linear dimensionality reduction, selecting the 2nd to 50th dimensions for UMAP visualization. We utilized the 'GeneActivity' function in Signac to compute gene activity in the scATAC-seq dataset by counting the chromatin fragments overlapping the gene body and a 2-kilobase upstream region for each gene. To generate a joint RNA/ATAC embedding, we calculate a WNN graph, representing a weighted combination of RNA and ATAC-seq modalities. We use this graph for UMAP visualization.

Peak calling
We imported ATAC-seq fragment data and GEX data into ArchR (v1.0.2) to construct an ArchR object, and subsequently added cell clustering and annotation information from Seurat object to the ArchR object. With the help of 'addReproduciblePeakSet' function, we then employed MACS2 to call peaks for each of the 13 cell types, with a fixed peak length of 500 bp. As a result, we called 251,078 unique ATAC peaks.

ChromVAR motif analysis
We used chromVAR (v.1.12.0) to quantify the enrichment of transcription factor (TF) motifs within accessible chromatin regions across single cells. Initially, we utilized the ArchR function 'addMotifAnnotations' to identify all matches of cisbp motifs within the peak set. Subsequently, we applied the 'addBgdPeaks' function to generate a set of genomic background peaks that were matched for copy number and accessibility. Finally, we calculated motif deviation z-scores for each cisbp motif using the 'addDeviationsMatrix' function.

Identification of differentially accessible chromatin regions
We used 'FindAllMarkers' function in Seurat to compute differentially accessible chromatin regions (DARs) for each cell type, with the main parameters: test.use = 'LR' and min.pct = 0.1. We obtained 42,747 DARs (|avg_logFC|> 0.25 and p_val <0.05). The top 4000 DARs were kept and used for further analysis. The 'FindMotifs' function was used to calculate motif enrichment for DARs of each cell type, allowing us to obtain cell-type-specific motifs. The 'Footprint' and 'PlotFootprint' functions were utilized to compute and visualize transcription factor footprints. The 'ClosestFeature' function was employed to identify the genes nearest to these peaks.

peak-to-gene linkage analysis
To find potential cis-regulatory elements (CRE) of genes, we used the 'LinkPeaks' function in Signac to calculate the Pearson correlation coefficient (correlation score >0.05 and p-value < 0.05) between gene expression and the accessibility of open chromatin regions, within 500 kb upstream and downstream of the transcription start site of each gene. The correlation scores greater than 0 indicated that the CRE might positively regulate gene transcription.

Dissecting candidate TFs driving differentiation trajectory
We utilized 'addTrajectory' function in ArchR to perform pseudotime ordering on epithelial and mesenchymal cells, with the epithelial lineage originating from Sox9$^+$ distal epithelia and the mesenchymal lineage originating from Wnt2$^+$ lung mes progenitors. Using 'getTrajectory' function, we extracted temporal features such as gene expression, chromatin accessibility, and motif chromVAR deviation scores. To identify TFs more specific to cell differentiation, we correlated these trajectories using the ArchR function 'correlateTrajectories' with default parameters. Finally, we identified transcription factors that exhibited both dynamic expression and dynamic chromatin enrichment during cellular developmental stages. These transcription factors are hypothesized to play roles in driving epithelial and mesenchymal differentiation.

Gene regulatory network analysis
We performed gene regulatory network analysis using SCENIC+ (v1.0a1) as described in the tutorial (https://scenicplus.readthedocs.io/en/latest/tutorials.html). Briefly, we began with data preparation. We extracted the gene expression matrix and converted it into the h5ad format, which was recognizable by Scanpy (v1.8.2). We also imported the ATAC fragments data into cisTopic (v2.0a0) to create a cisTopic object and appended the metadata derived from the Signac object. Subsequently, we ran the standard cisTopic pipeline to generate input files for SCENIC+. Additionally, as the tutorial recommended, we generated a custom cisTarget database for mm10 genome. After completing the preparatory work, we filled out the config.yaml file and ran the SCENIC+ workflow using SnakeMake (v8.5.5). Notably, we operated in the multiome mode. The primary results from the SCENIC+ analysis were stored in a MuData object. We retained the positively regulated eRegulons (rho_R2G > 0 and rho_TF2G > 0) for downstream analysis. Furthermore, we performed UMAP visualization of the cells based on the AUC values of the activator eRegulons, allowing cells belonging to the same cell type to cluster together. Using the AUC values calculated based on target genes and target regions, we utilized the 'heatmap_dotplot' function to display the clustering of all direct and extended eRegulons, respectively. We also created a network graph depicting the associations between selected transcription factors and their target regions and target genes using Cytoscape (v.3.10.3).

Statistics & reproducibility
Sample size was determined based on previous similar studies and statistical power calculations to ensure sufficient power to detect significant effects. No data exclusions were performed; all data points were included in the analysis. All experiments were replicated at least twice with consistent results, ensuring reproducibility. Data are presented as means ± SE in bar graphs. Data in box plots have center lines representing the dataset's median (50th percentile), with box bounds corresponding to the first quartile (25th percentile) and third quartile (75th percentile). Statistical significance analysis was performed using Wilcoxon rank-sum test or Student's t test for comparison between groups, as indicated in the figure legends. Levels of significance across all figures are indicated as follows: *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
Raw FASTQ data generated in this study are available at the NCBI Sequence Read Archive (SRA) under the reference number: PRJNA1203355 ("Single-cell Multiomics Analysis Reveals CTCF as a Key Regulator of Lung Morphogenesis and Progenitor Maintenance", [https://dataview.ncbi.nlm.nih.gov/object/PRJNA1203355]). All sequencing data generated in this study have been deposited in the Gene Expression Omnibus (GEO) database under accession number GSE286986. Data that supports the findings of this study are available in the main text, supplementary information, and source data. Source data are provided with this paper.

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Author contributions

S.S., X.T., and Y.J. conceived and designed the study; S.S., Y.J., and F.H. performed most of the experiments with the help of W.W., M.H., T.X., X.W., K.L., X.Z., Y.W., and H.H.; Y.J. and M.Y. performed the bioinformatics analysis; S.S., X.T., and X.L. supervised the work and acquired the funding support; and S.S., X.T., and Y.J. wrote the manuscript, with contributions from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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